

Big Data and Society

Class05 Information Cascade

**國企 Wen-Bin Chuang
2026-02-14**

Information Cascade 資訊級聯

An **information cascade** arises in a graph when nodes make adoption decisions **sequentially**, observing only the **actions** (adopted or not) of previous nodes, not their private signals. 資訊級聯是指在一個圖（graph）中，節點 **依次** 做出採納決策（adoption decision），**每個節點只能觀察到之前節點的行為（是否採納）**，而無法觀察到它們各自的**私有信號**。Classic theoretical model (Banerjee 1992, Bikhchandani-Hirshleifer-Welch 1992):

- Graph: usually assumed to be a countable infinite sequence (or sometimes a complete graph so everyone observes all previous actions). 通常假設為一個可數無限序列（或者有時假設為完全圖，使得每個人都能觀察到所有之前節點的行為）。
- Each node v has:
 - A private binary signal $s_v \in 0, 1$ (correct with probability $p > 0.5$). 一個私有二元信號 $s_v \in 0, 1$ （信號正確的機率 $p > 0.5$ ）
 - Observes the adoption decisions of all previous nodes. 可以觀察到所有先前節點做出的採納決策

- Node v adopts if the inferred Bayesian posterior (based on observed actions) exceeds 0.5, even if its own signal says "reject". 節點 v 的決策規則：如果根據所觀察到的行為推斷出的貝葉斯後驗概率 > 0.5 ，則採納該行為——即使自己的私有信號建議“拒絕”。

Result: With probability 1, after a finite number of steps, a cascade starts and **everyone ignores their private signal forever** → herd behavior. 結果：以概率 1，在有限步之後會出現資訊級聯（cascade），從此所有後續節點都永遠忽略自己的私有信號，形成從眾行為（herd behavior）。

Herd Behavior 從眾行為

A newly opened restaurant is actually of **poor** quality (true state = 0). However, each customer receives a **private signal** that is correct with probability (i.e., a 60% chance the signal is accurate and a 40% chance it is wrong). Customers arrive sequentially; they can only observe whether previous customers entered the restaurant, but they cannot see others' private signals. 一家新開的餐廳，真正品質其實是 **差** 的（真實狀態 = 0），但每個顧客都有一個私有信號，信號正確概率 $p = 0.6$ （即有60%概率信號正確，40%概率錯誤）。顧客依次到達，只能看到前面的人是否進去吃飯，看不到別人的私有信號。

Customer A trusts their own signal directly: since , they decide to enter. Starting from Customer B, even if their own private signal explicitly indicates "the restaurant is bad," the fact that they observe someone else having entered leads them to calculate a posterior probability still greater than 0.5. Consequently, they discard their own signal and follow the predecessor. A 直接相信自己信號： $0.6 > 0.5$ ，進去吃飯。從顧客 B 開始，即使他自己的信號明確告訴他“餐廳差”，但因為看到前面已經有人進去了，他計算出來的後驗概率仍然大於0.5，所以他放棄自己的信號，跟隨前面的人。

Once two consecutive customers enter (regardless of whether their decisions were right or wrong), the posterior beliefs of all subsequent customers become completely dominated by this public history of "2:0." No matter what their own private signals say, they will unconditionally follow suit. This marks the onset of an **information cascade**: from a certain point onward, all subsequent individuals permanently ignore their private information, falling into **herd behavior**. 一旦有兩個人連續都進去（無論對錯），後面所有顧客的後驗信念都會被徹底被“2:0”這個公開行為史支配，不管他們自己信號是什麼，都會無條件跟進。這就是**資訊級聯（資訊瀑布）**啟動了：從某一步開始，所有後續人都永遠忽略自己的私有資訊，陷入從眾（herd behavior）。

In this example, although the restaurant is actually poor, the first customer's incorrect signal ("good") triggers a cascade, causing the next 9,999 customers to all mistakenly enter. This results in a completely erroneous collective outcome, yet it is generated rationally step-by-step. This is the most classic and simplest case of an information cascade在這個例子中，餐廳其實是差的，但第1個顧客的錯誤信號（好）觸發了級聯，導致後面9999個顧客全部錯誤地進去吃飯——一個完全錯誤的群體行為，卻完全理性地逐個產生。這就是資訊級聯最經典、最簡單的案例。

Information Cascade = the real-world/social-science phenomenon of **herd behavior** caused by observational learning. 資訊級聯(Information Cascade) = 由觀察性學習引發的 **從眾行為** 這一現實世界/社會科學現象。

IC and LT models = the rigorous, scalable, graph-based computational models that capture and generalize exactly this phenomenon to arbitrary networks, making it possible to predict, maximize, or block cascades algorithmically. IC（獨立級聯）與 LT（線性閾值）模型 = 嚴謹且可擴展的、基於圖的計算模型，它們精確地捕捉並將這一現象推廣至任意網路結構，從而使得通過演算法預測、最大化或阻斷級聯成為可能。

Information Cascade is the herd behavior of real-world/social-science phenomenon, while graph-based computational diffusion models try to capture the phenomenon, making it possible to predict, maximize, or block cascades algorithmically. 簡而言之，**資訊級聯**是現實社會中的從眾現象，而**基於圖的計算擴散模型**則試圖對這一現象進行建模，以實現通過演算法對級聯進行預測、最大化或阻斷。

General Framework of Diffusion Models on Graphs

Independent Cascade (IC) Model 獨立級聯模型

The **Independent Cascade (IC) Model** is a **probabilistic model** used to simulate how information, influence, or innovation spreads through a network. 獨立級聯模型 是一種用於資訊、影響力或創新如何在網路中傳播的 **概率模型**。

- **The Core Idea:** Influence is like a contagious disease but with a twist. When an active node (someone who knows the info) tries to influence a neighbor, it gets **one single chance** to succeed. 影響力的傳播類似於傳染病，但有一個關鍵區別。當一個活躍節點（已知曉資訊的人）嘗試影響其鄰居時，它只有一**次機會**成功。
- **The Mechanism:** If the attempt succeeds (based on a probability), the neighbor becomes active in the *next* time step and gets their own chance to influence *their* neighbors. If the attempt fails, that specific link can never transmit influence again from that node. 如果嘗試成功（基於特定概率），鄰居將在**下一個時間步**變為活躍狀態，並獲得機會去影響它的鄰居。如果嘗試失敗，該特定連結將無法再次從該節點傳遞影響力。

- **The Process:** It starts with a set of **seed nodes** and runs in discrete steps until no new nodes can be activated. 模型從一組 **種子節點** 開始，以離散的時間步運行，直到沒有新節點可以被啟動為止。

Key Parameters

To run an IC simulation, you need three main components:

1. **The Graph:** A network where nodes represent individuals and edges represent social connections. 一個網路，其中節點代表個體，邊代表社會連接。
2. **Seed Nodes:** The initial set of active nodes (the "patient zero" or early adopters) at time . 初始活躍節點集合（即“零號病人”或早期採用者）。
3. **Propagation Probability** 傳播概率:
 - This is the probability that an active node successfully activates an inactive neighbor . 這是活躍節點 成功啟動其非活躍鄰居的概率。In the simplest version, is constant for all edges在最簡單的版本中， 對所有邊都是常數. In advanced versions, every edge can have its own specific probability . 在高級版本中，每條邊 可以擁有自己特定的概率 。

- **Herd Behavior:** The IC model mathematically formalizes the "Information Cascade". Even if a node has "private information" (represented here implicitly by not being connected to the seed), once enough neighbors activate, the probability of activation spikes, mimicking the rational decision to follow the crowd. IC 模型在數學上形式化了“資訊級聯”。即使一個節點擁有“私有資訊”（在此隱含地表現為未連接到種子節點），一旦足夠多的鄰居被啟動，其被啟動的概率就會激增，模仿了理性個體選擇跟隨大眾的行為。
- **Intervention:** By changing the `seeds` list, you can test strategies to **maximize** spread (Influence Maximization) or remove specific nodes/edges to **block** a cascade (e.g., stopping misinformation). 通過改變 seeds 列表，您可以測試**最大化**傳播的策略（影響力最大化），或者通過移除特定節點/邊來**阻斷**級聯（例如阻止虛假資訊傳播）

Linear Threshold (LT) Model 線性閾值模型

The **Linear Threshold (LT) Model** simulates spread based on **cumulative peer pressure** rather than a single probabilistic event. 線性閾值模型 是基於 **累積同伴壓力** (cumulative peer pressure) 的傳播過程，而非單一的隨機事件。

- **The Core Idea:** An individual adopts a behavior (or information) only when the **total influence** from their already-active neighbors exceeds a specific personal **threshold**. 個體只有在來自其已活躍鄰居的**總影響力**超過其特定的個人**閾值**時，才會採納某種行為（或資訊）。

- The Mechanism:
 1. Every node has a random **threshold**, usually between 0 and 1, representing its resistance to change. 每個節點都有一個隨機的**閾值**，通常在 0 到 1 之間，代表其對改變的抵抗力。
 2. Every incoming edge from an **active neighbor** contributes a certain "weight" of influence. 來自每個**活躍鄰居**的入邊都會貢獻一定的“權重”影響力。
 3. If the sum of weights from **active neighbors** the node's threshold, the node becomes active. 如果來自**活躍鄰居**的權重之和 該節點的閾值，該節點就會變為活躍狀態。
- **Key Difference from IC:** In IC, one strong connection might trigger activation instantly. In LT, it often takes **multiple** active neighbors to "push" a node over the edge. This models scenarios where people need social validation from several sources before acting. 在 IC 模型中，一個強連接可能瞬間觸發啟動。而在 LT 模型中，通常需要**多個**活躍鄰居共同作用，才能將節點“推過”臨界點。這類比了人們需要從多個來源獲得社會驗證後才會採取行動的場景。

Key Parameters

1. **The Graph:** The network structure. / **Seed Nodes :** The initial set of active nodes. 初始活躍節點集合。
 2. **Edge Weights:** The strength of influence from node to node 從節點 到 節點 的影響力強度。
 - Often normalized so that for any node , the sum of incoming weights equals 通常進行歸一化處理，使得對於任意節點，其所有入邊權重之和等於 1。
 3. **Node Thresholds** 節點閾值: A value assigned to each node (randomly drawn, e.g., from a uniform distribution). 分配給每個節點 的值（通常隨機抽取，例如服從均勻分佈）。
 - **Low :** Easy to influence (early adopter) 易受影響（早期採用者） / **High :** Resistant to influence (skeptic). 抗拒影響（懷疑論者）
- IC: **Probabilistic Push:** One neighbor tries to "infect" you 一個鄰居嘗試“感染”你. LT: **Cumulative Pull:** You activate when enough neighbors are active 當足夠多的鄰居活躍時，你被啟動

- **Institutional Herding:** The LT model perfectly captures the behavior of **fund managers** waiting for a **critical mass** of peers to hold a stock before they feel safe to enter (avoiding being the only one wrong). LT 模型完美捕捉了 **基金經理** 的行為——他們往往等待持有某檔股票的同行達到 **臨界多數** 後，才覺得安全並跟進（避免成為唯一犯錯的人）
- **Tipping Points:** The random threshold represents the specific internal risk tolerance or regulatory constraint of each investor. The model helps identify which small set of seeds can trigger a system-wide "regime shift." 隨機閾值代表了每個投資者內部特定的風險承受能力或監管約束。該模型有助於識別哪一小部分種子節點可以觸發系統性的“制度轉變”。
- **Blocking Cascades:** In LT, you can stop a cascade by increasing the thresholds of key nodes (e.g., via regulation or education) rather than just cutting links. 在 LT 模型中，您可以通過提高關鍵節點的閾值（例如通過監管或教育），而不僅僅是切斷連接，來阻止級聯的發生。這對於防範系統性風險尤為重要。

Triggering Model 觸發模型

The **Triggering Model** is a highly flexible framework where each node has a personal "**Triggering Set**" mechanism. 觸發模型是一個高度靈活的框架，其核心在於每個節點都擁有一個個性化的**觸發集合（Triggering Set）**機制。

- **The Core Idea:** Instead of a fixed probability (like IC) or a cumulative sum (like LT), every node randomly selects a **subset** of its incoming neighbors *at the very beginning* of the process. This subset is called its **Triggering Set**. 與 IC 模型的固定概率或 LT 模型的累積求和不同，在觸發模型中，每個節點在過程開始的**最初時刻**，會隨機選擇其入邊鄰居的一個**子集**。這個子集被稱為該節點的**觸發集合**。

- The Mechanism:
 1. **Selection Phase:** Before the simulation starts, every node flips a coin for each of its incoming neighbors to decide: "If *this* specific neighbor becomes active, will they be able to trigger me?" The neighbors that get a "Yes" form the Triggering Set. 在模擬開始前，每個節點 對其每一個入邊鄰居“擲硬幣”決定：“如果這個特定的鄰居變活躍了，他能否觸發我？”那些得到“是”的鄰居就構成了觸發集合
 2. **Activation Phase:** During the simulation, if **any single node** in 's Triggering Set becomes active, then immediately becomes active. 在過程中，只要 的觸發集合中有**任意一個**節點變為活躍狀態，節點 就會立即變為活躍狀態。

- **Key Difference:** It decouples the *probability of influence* from the *mechanism of activation*. It allows for complex correlations (e.g., "I only listen to Node A OR Node B, but not C"). 它將“影響力的概率”與“啟動的機制”解耦了。它允許複雜的邏輯關聯（例如：“我只聽命於節點 A **或** 節點 B，但不理會 C”）。
- Generalization:
 - If the triggering set contains at most one neighbor with probability , it behaves like **IC**. 如果觸發集合最多只包含一個鄰居（概率為 ），它的行為就像 **IC 模型**。
 - If the triggering set size follows a specific distribution based on weights, it can mimic **LT**. 如果觸發集合的大小遵循基於權重的特定分佈，它可以模仿 **LT 模型**。

Key Parameters

1. **The Graph:** Directed network (influence usually flows one way). 通常是有向網路（影響力通常是單向流動的）。
2. **Seed Nodes:** Initial active nodes. 初始活躍節點。
3. **Triggering Probabilities:** For every edge, there is a probability that is included in 's Triggering Set. 對於每條邊，存在一個概率表示被選入的觸發集合的可能性。In the standard version, these are independent Bernoulli trials. 在標準版本中，這些是獨立的伯努利試驗。
4. **The Rule:** Node activates if (i.e., if the intersection is not empty). 當且僅當（即交集非空，意味著觸發集合中至少有一個鄰居已活躍）時，節點啟動。

1. Institutional "Trust Networks":

- In finance, institutions don't just look at *anyone* holding a stock (LT) nor do they react to every single trade (IC). They have a specific **Watch List** or **Peer Group** (e.g., other ESG-focused funds, or specific aggressive investors). 在金融領域，機構不僅僅看任何人持有了某檔股票（LT 模型），也不對每一筆交易都做出反應（IC 模型）。他們有一個特定的**觀察名單 (Watch List)** 或 **同行組 (Peer Group)**（例如其他 ESG 專注基金，或特定的激進投資者）。
- Fund A only reacts if Fund B (whom they respect) enters the position. Fund C entering means nothing to Fund A. This maps directly to your interest in "**Strategic Holding by the Same Investor Group.**" 觸發模型完美捕捉了這一點：基金 A 只有在基金 B（他們尊重的對象）建倉時才會反應。基金 C 建倉對基金 A 毫無意義。這直接對應了您關注的**被相同投資者群體戰略性持有的概念**。

2. Filtering Noise:

- The "Triggering Set" acts as a filter. Even if 100 random people buy a stock, if none of them are in your specific Triggering Set, you remain inactive. This models **rational filtering** better than LT. 在觸發模型中，觸發集合充當了篩檢程式。即使有 100 個隨機人買入某檔股票，如果他們都不在你的特定觸發集合中，你依然保持不活躍。這比 LT 模型更好地類比了理性過濾行為。

3. Stealth Co-Accumulation:

- If a group of investors secretly coordinates, they effectively place each other in their Triggering Sets with probability 1.0, while ignoring the rest of the market. This creates a tightly coupled sub-graph that can trigger a cascade internally before leaking to the wider network. 如果一群投資者秘密協調，他們實際上是將彼此以 1.0 的概率放入對方的觸發集合中，而忽略市場的其餘部分。這會創建一個緊密耦合的子圖，能夠在內部觸發級聯，然後在洩露給更廣泛的網路之前完成建倉。

Influence Maximization (IM) 影響力最大化

Influence Maximization (IM) is the optimization problem of finding **the best set of "seed" nodes** to start a cascade, such that the total number of eventually activated nodes (the **spread**) is maximized. 影響力最大化是一個優化問題，旨在尋找最佳的種子節點集合，以啟動級聯傳播，從而使最終被啟動的節點總數（即傳播範圍）最大化。

- **The Core Question:** "If I can only convince people to start a rumor (or buy a product), *which* people should I choose to ensure the rumor reaches the maximum number of people?" “如果我只能說服一個人開始傳播謠言（或購買產品），我應該選擇哪個人，才能確保謠言觸達的人數最多？”
- **The Challenge:** In a network with millions of nodes, checking every possible combination of seeds is computationally impossible (it's an NP-hard problem). 在一個擁有數百萬節點的網路中，檢查個種子的每一種可能組合在計算上是不可行的（這是一個 **NP-hard** 問題）。

- **The Solution:** We rely on a mathematical property called **Submodularity** (the "diminishing returns" effect) to use a **Greedy Algorithm** that guarantees a solution within **63%** of the optimal result, which is often good enough for real-world applications. 我們利用一個稱為**次模性 (Submodularity)** 的數學性質（即“邊際收益遞減”效應），使用**貪心演算法 (Greedy Algorithm)**。該演算法能保證找到的解至少是最優解的 **63%**，這在現實應用中通常已經足夠好。

Key Concepts & Parameters

1. **Diffusion Model:** The underlying rules of spread (IC, LT, or Triggering) defined in previous steps. 前幾步定義的傳播規則（IC, LT 或 Triggering）
2. **Budget:** The number of seed nodes you are allowed to select. 允許選擇的種子節點數量
3. **Influence Spread:** The expected number of active nodes at the end of the process, given a seed set . Since IC/LT/Triggering are probabilistic, is usually estimated by running the simulation many times (Monte Carlo simulations) and averaging the results. 給定種子集合，過程結束時活躍節點的期望數量。由於 IC/LT/Triggering 是概率性的，通常通過運行多次模擬（蒙特卡洛模擬）並取平均值來估算。

4. **Submodularity:** This is the magic property. It means that adding a node to a *small* set of seeds gives more marginal gain than adding to a *large* set of seeds 這是關鍵屬性。意味著將一個節點 加入到一個小的種子集合中，其帶來的邊際增益要大於將其加入到一個大的集合中。
- *Analogy:* The first person you invite to a party adds a lot of value. The 100th person adds less value because most people already know someone there. 你邀請參加派對的第一個人價值巨大。第100個人的價值較小，因為大多數人已經認識派對上的其他人了。

5. **Strategic Intervention (ESG & Risk) 策略干預 (ESG 與風險):**

- **Maximizing:** If you want to promote a positive ESG narrative or a new sustainable investment strategy, IM tells you exactly which key institutions (hubs or bridges) to target first to trigger a market-wide shift. 如果您想推廣積極的 ESG 敘事或新的可持續投資策略，IM 會告訴您應該首先針對哪些關鍵機構（樞紐或橋樑節點），以觸發市場範圍的轉變。
- **Blocking (Negative IM):** Conversely, if you want to stop an "ESG run" or a panic sell-off, you can run IM on the *reverse* logic or identify the nodes whose removal minimizes the spread. These are your "systemically important" nodes for regulation. 反之，如果您想阻止“ESG 擠兌”或恐慌性拋售，您可以運行逆向邏輯的 IM，或者識別那些移除後能最小化傳播範圍的節點。這些就是您需要監管的“系統重要性”節點。

2. Identifying "Hidden" Influencers:

- Degree centrality (just counting connections) often fails. A node with moderate degree but positioned as a **bridge** between two dense clusters might be a better seed than a hub inside a single cluster. The Greedy IM algorithm naturally finds these **structural holes**. 度中心性（僅僅計算連接數）往往會失效。一個度中心性中等但作為兩個密集集群之間橋樑的節點，可能比單個集群內部的樞紐節點是更好的種子。貪心 IM 演算法能自然地發現這些 **結構洞** (Structural Holes)。

3. Cost-Benefit Analysis:

- The "Diminishing Returns" plot shows you when to stop spending resources. If the 4th seed adds very little extra spread compared to the 3rd, you know your budget is optimized at. “邊際收益遞減”圖表顯示了何時停止投入資源。如果第 4 個種子帶來的額外傳播範圍遠小於第 3 個，您就知道您的預算優化點在。

Reverse Influence Sampling (RIS) 反向影響力採樣

Greedy Algorithm was accurate but incredibly slow 雖然準確，但速度極慢 ($O(k \cdot N \cdot \text{Simulations})$).

- To pick just **one** seed, it had to simulate the spread for **every single node** in the network. 為了僅僅選擇一個種子節點，它必須對網路中的**每一個節點**都模擬一次傳播過程。
- For a network with 100,000 nodes, that's 100,000 simulations per step. It becomes computationally impossible for large-scale analysis. 對於一個擁有 10 萬個節點的網路，每一步都需要進行 10 萬次模擬。對於大規模分析來說，這在計算上是不可行的。
- **Greedy**: Accurate but too slow for big data 準確但對於大資料太慢. **RIS (TIM/IMM): Fast AND Accurate.** This is the industry standard for large-scale influence maximization today. **既快又准**。這是當今大規模影響力最大化的行業標準。

Easy Definition of RIS

Reverse Influence Sampling (RIS) flips the logic upside down. Instead of asking "*If I start at Node A, how far does it spread?*", 將邏輯完全顛倒過來。它不再問“如果我從節點 A 開始，能傳播多遠？”

It asks: **If a random node v ended up activated, which seeds could have caused it?** 如果一個隨機節點 v 最終被啟動了，那麼是哪些種子節點可能導致了它的啟動？

- The Core Idea: Generate many "Reverse Reachable Sets" (RR-sets). 生成大量的**反向可達集合** (Reverse Reachable Sets, RR-sets)
 1. Pick a random node v from the graph.
 2. Simulate the diffusion process **backwards** from v . 從圖中隨機選擇一個節點 v
 3. Collect all nodes that *could* have reached v in this specific random realization. This group is an **RR-set**. 從 v 開始**反向**模擬擴散過程，收集在這個特定的隨機實現中**能夠到達** v 的所有節點。這個群體就是一個 **RR-set**.
- **The Insight:** If a node u appears in **many** different RR-sets, it means u has a high probability of influencing a large portion of the network. 如果一個節點 u 出現在**許多**不同的 RR-set 中，這意味著 u 有很高的概率去影響網路中的大部分節點。

- The Algorithm:
 1. Generate enough RR-sets to cover the graph statistically. 生成足夠多的 RR-sets，以在統計上覆蓋整個圖
 2. Solve a **Maximum Coverage Problem**: Pick k nodes that appear in the *maximum number* of RR-sets. 求解**最大覆蓋問題**（******Maximum Coverage Problem**）：選擇 k 個出現在*最多 RR-sets 中的節點。
 3. This can be solved extremely fast using a simple greedy approach on the sets, not the graph. 這可以通過在集合上運行簡單的貪心演算法來極快地解決，而不需要再對圖進行昂貴的模擬。

1. Scalability to Financial Networks 金融網路的可擴展性:

- When analyzing networks with **100,000+ nodes** (e.g., `watts.strogatz.game(100000)`), the standard Greedy algorithm would take days or weeks to run on such data. 標準的貪心演算法在這樣的資料上運行可能需要數天甚至數周。
- **RIS** can solve the same problem in **minutes or seconds**. This makes it feasible to run daily risk assessments or real-time strategy simulations on full-market data. **RIS** 可以在**幾分鐘甚至幾秒鐘**內解決同樣的問題。這使得在完整的市場資料上進行每日風險評估或即時策略模擬成為可能。

2. Identifying Systemic Risk Sources 識別系統性風險源:

- In the context of "**ESG Run Risks**" or "**Stealth Co-Accumulation**", you don't just want to know who has the most connections (Degree). You want to know who has the highest probability of triggering a cascade across the *entire* system. 在“**ESG 擠兌風險 或 安靜協同建倉**”的背景下，您不僅僅想知道誰的連接最多（度數）。您想知道誰最有概率觸發貫穿整個系統的級聯。
- Nodes that appear frequently in RR-sets are the true "systemic triggers." These are the institutions whose failure or strategic shift would ripple through the maximum number of other funds. 頻繁出現在 RR-sets 中的節點是真正的“系統性觸發器”。這些機構一旦失敗或發生戰略轉變，其漣漪效應將波及最多的其他基金。

3. Robustness:

- Because RIS relies on statistical sampling (Law of Large Numbers), it is robust against noise in the network structure. It averages out the randomness of individual edges, giving a clearer picture of structural importance. 由於 RIS 依賴於統計採樣（大數定律），它對網路結構中的噪音具有魯棒性。它平均掉了單個邊的隨機性，提供了更清晰的結構重要性圖景。

Minimizing Spread 最小化傳播

It is also known as **Negative Influence Maximization** or **Cascade Blocking**). This is the critical defensive counterpart to Influence Maximization, essential for risk management. **負面影響力最大化或級聯阻斷**（Cascade Blocking）。這是影響力最大化的關鍵防禦對應面，對於風險管理至關重要。

Minimizing Spread is the optimization problem of finding the best set of nodes to **remove**, **immunize**, or **counter-act** in order to stop a negative cascade (e.g., panic selling, misinformation, ESG run) from reaching as many people as possible. **最小化傳播**是一個優化問題，旨在尋找最佳的節點集合進行**移除**、**免疫**或**反制**，以阻止負面級聯（如恐慌性拋售、虛假資訊、ESG 擠兌）波及盡可能多的節點。

- **The Core Question:** "If a crisis starts (or has already started), which k nodes should we intervene with (by removing them, adding firewalls, or injecting positive news) to save the maximum number of other nodes?" 如果危機已經開始（或即將開始），我們應該干預哪 k 個節點（通過移除它們、建立防火牆或注入正面資訊），以挽救最大數量的其他節點？

- The Strategies:
 1. **Node Removal (Immunization):** Completely disconnecting key nodes (e.g., halting trading for specific institutions, isolating a server). **節點移除（免疫）：**完全斷開關鍵節點的連接（例如，暫停特定機構的交易，隔離伺服器）。
 2. **Positive Counter-Cascade:** Injecting a competing "positive" signal (e.g., official reassurance, buy orders) that competes with the negative one. If a node hears the positive signal first (or from a more trusted source), it becomes immune to the negative one. **正面反級聯（Positive Counter-Cascade）：**注入競爭的“正面”信號（例如，官方安撫、買入指令），與負面信號競爭。如果一個節點先接收到正面信號（或來自更可信的來源），它將對負面信號免疫。
- **The Challenge:** Like maximization, finding the *optimal* set is NP-hard. However, the objective function (number of saved nodes) is often **supermodular** (or the reduction in spread is submodular), allowing us to use similar Greedy or RIS-based algorithms with theoretical guarantees. 與最大化問題一樣，尋找最優集合是 NP-hard 問題。然而，目標函數（被挽救的節點數）通常具有**超模性（supermodular）**（或者傳播減少量具有次模性），這使得我們可以使用類似的貪心演算法或基於 RIS 的演算法，並獲得理論保證。

Key Concepts & Parameters

1. **Threat Source (S_{bad}):** The initial set of nodes spreading the negative influence (e.g., the first funds to sell). 傳播負面影響的初始節點集合（例如，第一批拋售的基金）。
2. **Budget (k):** The number of nodes you can protect/remove or the number of positive seeds you can plant. 您可以保護/移除的節點數量，或可以種植的正面種子數量。
3. Competing Models:
 - In a **Competing IC/LT Model**, if a node is activated by the "Positive" campaign before the "Negative" one reaches it, the node is saved. 在**競爭 IC/LT 模型**中，如果節點在“負面”浪潮到達之前被“正面”活動啟動，則該節點被挽救。
 - Alternatively, removing a node reduces the network's connectivity, breaking the paths of the negative cascade. 或者，移除節點會降低網路的連通性，切斷負面級聯的路徑。
4. **Objective:** Minimize $\sigma(S_{\text{bad}})$ (the final size of the bad cascade) or Maximize $\text{Saved} = N - \sigma(S_{\text{bad}})$. 負面級聯的最終規模）或最大化 $\text{Saved} = N - \sigma(S_{\text{bad}})$ （被挽救的節點數）

1. Targeting Bridges vs. Hubs: 橋樑 vs. 樞紐

- In many cases, the algorithm will choose to remove the **seed itself** (if allowed) as the first step. This is the most obvious "cure." 在許多情況下，演算法第一步就會選擇移除**種子節點本身**（如果允許的話）。這是最直接的“治癒”方法。
- If the seed cannot be removed (e.g., the crisis has already started inside a bank you can't close), the algorithm will likely target **bridges**—nodes that connect the infected cluster to the rest of the healthy network. This effectively quarantines the outbreak. 如果種子無法被移除（例如危機已經發生在一家不能關閉的銀行內部），演算法通常會瞄準**橋樑節點**——那些連接受感染集群與健康網路其餘部分的節點。這有效地將疫情**隔離**在局部。

2. Application to "ESG Runs":

- **Scenario:** A rumor starts that Fund X is greenwashing. 謠言開始傳播，稱基金 X 存在“漂綠”行為。
- **Action:** The regulator identifies the top 3 funds that act as "transmitters" (high connectivity to other susceptible funds) and temporarily suspends their ability to trade or issues a strong clarifying statement specifically targeted at their immediate neighbors to "cut the link." 監管機構識別出前 3 個充當“傳播者”的基金（與其他易感基金連接度高），並暫時暫停其交易能力，或向其直接鄰居發佈針對性的澄清聲明以“切斷連結”。
- **Result:** The rumor dies out locally instead of becoming a market-wide panic. 謠言在局部消亡，而不是演變成全市場的恐慌。

3. Application to "Systemic Risk":

- This models the concept of "**Too Connected to Fail**" in reverse. Instead of bailouts, it identifies which institutions represent such a high propagation risk that they require strict **firewalls** (capital controls, trading halts) to protect the broader system. 這反向模擬了**“大而不能倒” (Too Connected to Fail)** 的概念。與其進行救助，不如識別哪些機構代表了極高的傳播風險，因此需要嚴格的**防火牆**（資本管制、交易暫停）來保護更廣泛的系統。